

Dimensional Analysis and Significant Figures

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Part I

Dimensional Analysis

Problem Set

Chapter 1

JEE Mains Problems

1.1 Intext Problems

Q1: The dimensional formula of $\frac{1}{2}\epsilon_0 E^2$ (ϵ_0 = permittivity of vacuum and E = Electric Field) is $M^a L^b T^c$. The value of $2a - b + c = 0$ is _____.

[JEE Mains 2nd April 2026 Shift 1]

- A) 0
- B) 1
- C) -1
- D) 2

{Solution : <https://www.youtube.com/watch?v=CAgTYRH7-P8> }

1.2 Full Syllabus Problems

Q1: The position of an object having mass 0.1 kg as a function of time t is given as $\vec{r} = 10t^2\hat{i} + 5t^3\hat{j}$ m. At $t = 1$ s, which of the following statements is correct ?

[JEE Mains 2nd April 2026 Shift 1]

- A The linear momentum $\vec{p} = (2\hat{i} + 1.5\hat{j})kg.m/s$.
- B The force acting on the object $\vec{F} = (2\hat{i} + 3\hat{j}) N$.
- C The angular momentum of the object about it's origin $\vec{L} = 15\hat{k} J.s$.
- D The torque acting on the object about it's origin $T = 20\hat{k} N.m$.

Choose the correct answer from the options given below :

- (a) A, B and C only.
- (b) B, C and D only.
- (c) A, C and D only.
- (d) A, B and D only.

{Solution : <https://www.youtube.com/watch?v=DvIERvObc1E>}

Part II

Significant Figures

Problem Set

Chapter 2

JEE Mains Problems

Q1 : The diameter of a wire measured by a screw gauge of least count 0.001 cm is 0.08 cm. The length measured by a scale of least count 0.1 cm is 150 cm. When a weight of 100 N is applied to the wire, the extension in length is 0.5 cm, measured by a micrometer of least count 0.001 cm. The error in the measurement of Young's Modulus is $\alpha \times 10^9 \text{ N/m}^2$. The value of α is

(Ignore the contribution of load to Young's Modulus error calculation)
[JEE Mains 2nd April 2026 Shift 1]
A) 1.3
B) 1.65
C) 0.13
D) 0.25
{Solution : <https://www.youtube.com/watch?v=JCaMvvfacDw>}